United States Patent Application Publication	20250275329
Kind Code	A1
Publication Date	August 28, 2025
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LIGHT EMITTING ELEMENT

Abstract

A light emitting element includes a semiconductor stack structure, a first electrode, a second electrode, and an insulation layer. The semiconductor stack structure includes a first p-type semiconductor layer, a first active layer, a first n-type semiconductor layer, an intermediate layer, a second p-type semiconductor layer, a second active layer, and a second n-type semiconductor layer. The semiconductor stack structure includes a first opening and a second opening. The first opening is provided continuously in the first p-type semiconductor layer and the first active layer. The second opening in a plan view is located to overlap the first opening. The second opening is provided continuously in the first n-type semiconductor layer, the intermediate layer, the second p-type semiconductor layer, and the second active layer.

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Appl. No.:	**19/061751**
Filed:	**February 24, 2025**

Foreign Application Priority Data

JP	2024-027681	Feb. 27, 2024

Publication Classification

Int. Cl.: **H10H29/14** (20250101); **H10H29/01** (20250101); **H10H29/80** (20250101); **H10H29/85** (20250101)

U.S. Cl.:

CPC **H10H29/142** (20250101); **H10H29/011** (20250101); **H10H29/032** (20250101); **H10H29/034** (20250101); **H10H29/8321** (20250101); **H10H29/842** (20250101); **H10H29/857** (20250101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-027681, filed on Feb. 27, 2024, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] An embodiment of the present disclosure relates to a light emitting element.

[0003] U.S. Patent Publication No. 2009/0001389 discloses a structure in which a first LED including a green light emitting first active region and a second LED including a blue light emitting second active region are stacked via a tunnel junction.

SUMMARY

[0004] An object of an embodiment of the present disclosure is to provide a light emitting element capable of reducing light distribution variation between the first active layer and the second active layer.

[0005] A light emitting element according to one embodiment of the present

[0006] disclosure has a semiconductor stack structure, a first electrode, a second electrode, and an insulation layer. The semiconductor stack structure has a first p-type semiconductor layer, a first active layer, a first n-type semiconductor layer, an intermediate layer, a second p-type semiconductor layer, a second active layer, and a second n-type semiconductor layer. The first active layer is disposed on the first p-type semiconductor layer. The first n-type semiconductor layer is disposed on the first active layer. The intermediate layer is disposed on the first n-type semiconductor layer. The second p-type semiconductor layer is disposed on the intermediate layer. The second active layer is disposed on the second p-type semiconductor layer. The second n-type semiconductor layer is disposed on the second active layer. The semiconductor stack structure has a first opening and a second opening. The first opening is provided continuously in the first p-type semiconductor layer and the first active layer. The second opening in a plan view is located to overlap the first opening. The second opening is provided continuously in the first n-type semiconductor layer, the intermediate layer, the second p-type semiconductor layer, and the second active layer. The first electrode is located in the first opening. The first electrode is electrically connected to the first n-type semiconductor layer. The second electrode is located in the first opening and the second opening. The second electrode is electrically connected to the second n-type semiconductor layer. The insulation layer is disposed between the first electrode and the second electrode. The insulation layer is in contact with the first electrode and the second electrode.

[0007] According to an embodiment of the present disclosure, a light emitting element capable of reducing light distribution variation between the first active layer and the second active layer can be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a plan view showing a light emitting element according to a first embodiment.

[0009] FIG. **2** is a cross-sectional view of the light emitting element according to the first embodiment.

[0010] FIG. **3** is a cross-sectional view of the current path in a first emission state of the light emitting element according to the first embodiment.
[0011] FIG. **4** is a cross-sectional view of the current path in a second emission state of the light emitting element according to the first embodiment.
[0012] FIG. **5** is a cross-sectional view of the current path in a third emission state of the light emitting element according to the first embodiment.
[0013] FIG. **6** is a plan view of a light emitting element according to a variation of the first embodiment.
[0014] FIG. **7** is a cross-sectional view of the light emitting element according to the variation of the first embodiment.
[0015] FIG. **8** is a plan view of a light emitting element according to a second embodiment.
[0016] FIG. **9** is a cross-sectional view of the light emitting element according to the second embodiment.
[0017] FIG. **10** is a plan view of a light emitting element according to a variation of the second embodiment.
[0018] FIG. **11** is a plan view of a light emitting element according to a third embodiment.
[0019] FIG. **12** is a plan view of a light emitting element according to a variation of the third embodiment.
[0020] FIG. **13** is a cross-sectional view showing a part of a method of manufacturing a light emitting element according to the first embodiment.
[0021] FIG. **14** is a cross-sectional view showing a part of the method of manufacturing a light emitting element according to the first embodiment.
[0022] FIG. **15** is a cross-sectional view showing a part of the method of manufacturing a light emitting element according to the first embodiment.
[0023] FIG. **16** is a cross-sectional view showing a part of the method of manufacturing a light emitting element according to the first embodiment.
[0024] FIG. **17** is a cross-sectional view showing a part of the method of manufacturing a light emitting element according to the first embodiment.

EMBODIMENTS

[0025] Certain embodiments of the present disclosure will be described below with reference to the accompanying drawings.
[0026] The drawings are schematic or conceptual. Consequently, the relationships of the thicknesses and the widths between the members or the size ratio of the members might not necessarily be the same as those in reality. Even when showing the same portion, the dimensions of or the ratio between the members might be different depending on the drawing.
[0027] In the present specification and the drawings, the elements that are similar to those which have been described using a drawing previously referenced are denoted by the same reference numerals for which detailed explanation will be omitted as appropriate.
[0028] In the description below, an XYZ orthogonal coordinate system is employed to explain the layout and the structure of each part or portion for explanation purposes. The X-axis, Y-axis, and Z-axis are orthogonal to one another. The direction in which the X-axis extends is designated as “X direction,” and the direction in which the Y-axis extends is designated as “Y direction,” and the direction in which the Z-axis extends is designated as “Z direction.” In the Z direction, the direction indicated by the arrow may be referred to as upward and the direction opposite thereto may also be referred to as downward to make the explanation more easily understood, but these directions are irrespective of the direction of gravity. A “plan view” refers to a view when viewing an object along the Z direction.

First Embodiment

[0029] FIG. **1** is a plan view showing a light emitting element according to a first embodiment.
[0030] FIG. **2** is a cross-sectional view of the light emitting element according to the first

embodiment.
[0031] FIG. **2** is a cross section taken along line II-II in FIG. **1**.
[0032] As shown in FIG. **1** and FIG. **2**, a light emitting element **100** according to the first embodiment includes a semiconductor stack structure **10**, a first pad electrode **21**, a second pad electrode **22**, a lower electrode **23**, a first electrode **31**, a second electrode **32**, a connection electrode **53**, and an insulation layer **45**.
[0033] As shown in FIG. **1**, the plan view shape of the light emitting element **100** is quadrangular. In the case in which the plan view shape of the light emitting element **100** is quadrangular, the length of a side of the quadrangle is, for example, 50 μm to 2000 μm.
[0034] The semiconductor stack structure **10** has a first p-type semiconductor layer **11***a*, a first active layer **11***b*, a first n-type semiconductor layer **11***c*, an intermediate layer **13**, a second p-type semiconductor layer **12***a*, a second active layer **12***b*, and a second n-type semiconductor layer **12***c*. The first active layer **11***b*, the first n-type semiconductor layer **11***c*, the intermediate layer **13**, the second p-type semiconductor layer **12***a*, and the second active layer **12***b* are positioned between the first p-type semiconductor layer **11***a* and the second n-type semiconductor layer **12***c*.
[0035] The first p-type semiconductor layer **11***a* is disposed at the lowermost portion of the semiconductor stack structure **10**. The first active layer **11***b* is disposed on the first p-type semiconductor layer **11***a*. The first n-type semiconductor layer **11***c* is disposed on the first active layer **11***b*. The intermediate layer **13** is disposed on the first n-type semiconductor layer **11***c*. The second p-type semiconductor layer **12***a* is disposed on the intermediate layer **13**. The second active layer **12***b* is disposed on the second p-type semiconductor layer **12***a*. The second n-type semiconductor layer **12***c* is disposed on the second active layer **12***b*.
[0036] The first p-type semiconductor layer **11***a*, the first active layer **11***b*, the first n-type semiconductor layer **11***c*, the intermediate layer **13**, the second p-type semiconductor layer **12***a*, the second active layer **12***b*, and the second n-type semiconductor layer **12***c* are each formed of a nitride semiconductor, for example. In the present specification, “nitride semiconductors” include semiconductors of all compositions obtained by varying the composition ratio x and y within their ranges in the chemical formula $In_xAl_yGa_{1-x-y}N$ ($0 \le x \le 1$, $0 \le y \le 1$, $x+y \le 1$). Moreover, those further containing a group V element in addition to nitrogen (N), and those further containing various elements added for controlling various physical properties such as conductivity type are also included in “nitride semiconductors.”
[0037] The first n-type semiconductor layer **11***c* and the second n-type semiconductor layer **12***c* contain, for example, Si (silicon) as an n-type impurity. The first p-type semiconductor layer **11***a* and the second p-type semiconductor layer **12***a* contain, for example, Mg (magnesium) as a p-type impurity. The first active layer **11***b* and the second active layer **12***b* are emission layers that emit light and have, for example, a MQW (multiple quantum well) structure including multiple barrier layers and multiple well layers. The peak wavelength of the light emitted by the first active layer **11***b* may be the same as or different from the peak wavelength of the light emitted by the second active layer **12***b*.
[0038] The intermediate layer **13** includes at least either a p-type semiconductor layer having a higher p-type impurity concentration than that of the second p-type semiconductor layer **12***a* or an n-type semiconductor layer having a higher n-type impurity concentration than that of the first n-type semiconductor layer **11***c*. The intermediate layer **13** functions as a tunnel junction layer, for example.
[0039] In the light emitting element **100**, the second active layer **12***b* is larger in area than the first active layer **11***b* in a plan view.
[0040] The semiconductor stack structure **10** has a first opening **16** and a second opening **17**. The first opening **16** is provided continuously in the first p-type semiconductor layer **11***a* and the first active layer **11***b*. The first opening **16** is a hole that passes through the first p-type semiconductor layer **11***a* and the first active layer **11***b* in the Z direction. In the light emitting element **100**, the first

opening **16** is provided continuously in the first p-type semiconductor layer **11**_a_, the first active layer **11**_b_, and the first n-type semiconductor layer **11**_c_. In the light emitting element **100**, the first opening **16** is a hole that passes through the first p-type semiconductor layer **11**_a_ and the first active layer **11**_b_ before reaching the first n-type semiconductor layer **11**_c_ in the Z direction. The first opening **16** does not pass through the first n-type semiconductor layer **11**_c_. In the light emitting element **100**, the semiconductor stack structure **10** has multiple first openings **16**. The semiconductor stack structure **10** has at least one first opening **16**.

[0041] The second opening **17** is provided continuously in the first p-type semiconductor layer **11**_a_, the first active layer **11**_b_, the first n-type semiconductor layer **11**_c_, the intermediate layer **13**, the second p-type semiconductor layer **12**_a_, and the second active layer **12**_b_. The second opening **17** is a hole that passes through the first p-type semiconductor layer **11**_a_, the first active layer **11**_b_, the first n-type semiconductor layer **11**_c_, the intermediate layer **13**, the second p-type semiconductor layer **12**_a_, and the second active layer **12**_b_. In the light emitting element **100**, the second opening **17** is continuously provided in the first p-type semiconductor layer **11**_a_, the first active layer **11**_b_, the first n-type semiconductor layer **11**_c_, the intermediate layer **13**, the second p-type semiconductor layer **12**_a_, the second active layer **12**_b_, and the second n-type semiconductor layer **12**_c_. In the light emitting element **100**, the second opening **17** is a hole that passes through the first p-type semiconductor layer **11**_a_, the first active layer **11**_b_, the first n-type semiconductor layer **11**_c_, the intermediate layer **13**, the second p-type semiconductor layer **12**_a_, and the second active layer **12**_b_ before reaching the second n-type semiconductor layer **12**_c_ in the Z direction. The second opening **17** does not pass through the second n-type semiconductor layer **12**_c_. In the light emitting element **100**, the semiconductor stack structure **10** has multiple second openings **17**. The semiconductor stack structure **10** has at least one second opening **17**. The second opening **17** is provided at a location that overlaps the first opening **16** in a plan view.

[0042] In the light emitting element **100**, the plan view shape of each of the first openings **16** and the second openings **17** is a perfect circle. The plan view shape of each of the first openings **16** and the second openings **17** may be a semicircle, ellipse, polygon, or the like. In the case in which the shape of each of the first openings **16** and the second openings **17** is a perfect circle, the diameter of a first opening **16** is, for example, 15 μm or smaller, preferably 10 μm to 15 μm, and the diameter of a second opening **17** is, for example, smaller than 12 μm, preferably 5 μm to 10 μm.

[0043] In the light emitting element **100**, the second connection part **32**_a_ where the second electrode **32** is in contact with the second n-type semiconductor layer **12**_c_ in a second opening **17** is surrounded by the first connection part **31**_a_ where the first electrode **31** is in contact with the first n-type semiconductor layer **11**_c_ in the first opening **16** that overlaps the second opening **17** in a plan view.

[0044] At least a portion of the upper face of the semiconductor stack structure **10** has multiple projections, for example. As shown in FIG. **2**, the projections are arranged to overlap the first active layer **11**_b_ and the second active layer **12**_b_ in the Z direction. The projections provided on the upper face of the semiconductor stack structure **10** can increase the extraction efficiency of the light emitted upwards from the semiconductor stack structure **10**. The upper face of the semiconductor stack structure **10** is the upper face of the second n-type semiconductor layer **12**_c_. The projections may be provided not only in the region that overlaps both the first active layer **11**_b_ and the second active layer **12**_b_, but also in the region that overlaps only the second active layer **12**_b_.

[0045] The first pad electrode **21** is electrically connected to the first p-type semiconductor layer **11**_a_. For example, the first pad electrode **21** is positioned apart from the semiconductor stack structure **10** in a plan view. The first pad electrode **21** does not overlap the semiconductor stack structure **10** in the Z direction, for example. The first pad electrode **21** may overlap the semiconductor stack structure **10** in the Z direction.

[0046] The second pad electrode **22** is electrically connected to the first n-type semiconductor layer

11*c*. For example, the second pad electrode **22** is positioned apart from the semiconductor stack structure **10** in a plan view. In the example of FIG. **2**, the second pad electrode **22** does not overlap the semiconductor stack structure **10** in the Z direction. However, the second pad electrode **22** may overlap the semiconductor stack structure **10** in the Z direction. For example, the second pad electrode **22** may be provided at a location opposite the first pad electrode **21** with the semiconductor stack structure **10** interposed between the second pad electrode **22** and the first pad electrode **21** in a plan view.

[0047] The lower electrode **23** is electrically connected to the second n-type semiconductor layer **12***c*. The lower electrode **23** is provided under the semiconductor stack structure **10**, for example. In the light emitting element **100**, the lower electrode **23** is disposed at the lowermost portion of the light emitting element **100**. In the light emitting element **100**, the lower electrode **23** is placed under the semiconductor stack structure **10**, the first pad electrode **21**, the second pad electrode **22**, the first electrode **31**, the second electrode **32**, the connection electrode **35**, and the insulation layer **45**. The light emitting element **100** is mounted on a substrate or the like by bonding the lower electrode **23** and the substrate with a bonding material, such as solder.

[0048] The first pad electrode **21** and the second pad electrode **22** are each formed of a metal material, for example. For the metal material, at least one of metals, such as Ti (titanium), Pt (platinum), Rh (rhodium), Au (gold), Ni (nickel), Ta (tantalum), Zr (zirconium), and alloys containing these metals can be used. The first pad electrode **21** and the second pad electrode **22** may each be a single layer, or a multilayer structure in which multiple layers are stacked. The first pad electrode **21** and the second pad electrode **22** can each be a multilayer structure in which a Ti layer, a Pt layer, and an Au layer are stacked in that order, for example.

[0049] The lower electrode **23** is formed of at least either a metal material or semiconductor material, for example. For the metal material, the same metal material used for the first pad electrode **21** and the second pad electrode **22** can be used. For the semiconductor material, for example, Si can be used. The lower electrode **23** may be, for example, a multilayer structure in which a layer formed of a semiconductor material is stacked on a metal layer.

[0050] The first electrode **31** electrically connects the second pad electrode **22** and the first n-type semiconductor layer **11***c*. The first electrode **31** is located in the first openings **16**. The first electrode **31** is not located in the second openings **17**. In the light emitting element **100**, the first electrode **31** extends from the area under the second pad electrode **22** to the area under the first n-type semiconductor layer **11***c*. In the light emitting element **100**, the first electrode **31** and the first n-type semiconductor layer **11***c* are electrically connected by coming into contact with one another at the first connection parts **31***a*.

[0051] The second electrode **32** electrically connects the lower electrode **32** and the second n-type semiconductor layer **12***c*. The second electrode **32** is located in the first openings **16** and the second openings **17**. In the light emitting element **100**, the second electrode **32** is provided under the second n-type semiconductor layer **12***c*. In the light emitting element **100**, the lower electrode **23** and the second n-type semiconductor layer **12***c* are electrically connected as the second electrode **32** and the second n-type semiconductor layer **12***c* come into contact with one another at the second connection parts **32***a*.

[0052] The connection electrode **35** electrically connects the first pad electrode **21** and the first p-type semiconductor layer **11***a*. In the light emitting element **100**, the connection electrode **35** extends from the area under the first pad electrode **21** to the area under the first p-type semiconductor layer **11***a*. In the light emitting element **100**, the connection electrode **35** and the first p-type semiconductor layer **11***a* are electrically connected via the interlayer electrode **25** positioned between the connection electrode **35** and the first p-type semiconductor layer **11***a*.

[0053] The first electrode **31**, the second electrode **32**, and the connection electrode **35** are each formed of a metal material, for example. For the metal material, for example, at least one of metals, such as Au (gold), Pt (platinum), Pd (palladium), Rh (rhodium), Ni (nickel), W (tungsten), Mo (molybdenum), Cr (chromium), Ti (titanium), Al (aluminum), Cu (copper), In (indium), Pb (zinc), and alloys containing these metals can be used. The first electrode **31**, the second electrode **32**, and the connection electrode **35** may each be a single layer, or a stack structure in which multiple layers are stacked. The first electrode **31**, the second electrode **32**, and the connection electrode **35** may each be a stack structure in which a Ti layer, an Al alloy layer, a Ti layer, a Pt layer, an Au layer, and a Ti layer are stacked in that order, for example.

[0054] The insulation layer **45** is disposed between the first electrode **31** and the second electrode **32**. The insulation layer **45** electrically isolates the first electrode **31** from the second electrode **32**.

[0055] The insulation layer **45** is formed of an insulation material. For the insulation material, for example, at least one of silicon oxide, silicon nitride, and silicon oxynitride can be used.

[0056] The light emitting element **100** further includes an interlayer insulation film **50**. The interlayer insulation film **50** is disposed between the semiconductor stack structure **10** and the first electrode **31**, between the semiconductor stack structure **10** and the second electrode **32**, and between the semiconductor stack structure **10** and the connection electrode **35**. A portion of the interlayer insulation film **50** is disposed between the semiconductor stack structure **10** and the second electrode **32** in a first opening **16**. The portion of the interlayer insulation film **50** surrounds the first electrode **31** in a plan view. Another portion of the interlayer insulation film **50** is disposed between the first electrode **31** and the connection electrode **35**, electrically isolating the first electrode **31** from the connection electrode **35**. Another portion of the interlayer insulation film **50** is disposed between the second electrode **32** and the connection electrode **35**, electrically isolating the second electrode **32** from the connection electrode **35**.

[0057] The interlayer insulation film **50** is formed of an insulation material, for example. For the insulation material, for example, at least one of silicon oxide, silicon nitride, and silicon oxynitride can be used.

[0058] The light emitting element **100** further incudes a protective film **55**. The protective film **55** is disposed on the semiconductor stack structure **10**. The protective film **55** protects the semiconductor stack structure **10**. For example, the thickness of the protective film **55** at the location that overlaps the projections of the upper face of the semiconductor stack structure **10** is preferably smaller than the thickness of the protective film **55** located elsewhere. This can increase the extraction efficiency of the light emitted upwards by the semiconductor stack structure **10**.

[0059] The protective film **55** is formed of an insulation material, for example. For the insulation material, for example, an oxide or nitride containing at least one selected from the group consisting of Si (silicon), Ti (titanium), Zr (zirconium), Nb (niobium), Ta (tantalum), Al (aluminum), and Hf (hafnium) can be used. For the insulation material, for example, at least one of silicon oxide, silicon nitride, and silicon oxynitride can be used.

[0060] As shown in FIG. **1** and FIG. **2**, in the light emitting element **100**, the contact area between the first electrode **31** and the first n-type semiconductor layer **11***c* in a first opening **16** is larger than the contact area between the second electrode **32** and the second n-type semiconductor layer **12***c* in a second opening **17** in a plan view. The contact area between the second electrode **32** and the second n-type semiconductor layer **12***c* in a second opening **17** is preferably 10% or more, but less than 100% of the contact area between the first electrode **31** and the first n-type semiconductor layer **11***c* in a first opening **16**, for example. In the case in which the semiconductor stack structure **10** has multiple first openings **16** and multiple second openings **17**, the sum of the contact areas between the first electrode **31** and the first n-type semiconductor layer **11***c* in the first openings **16** is preferably larger than the sum of the contact areas between the second electrode **32** and the second n-type semiconductor layer **12***c* in the second openings **17** in a plan view. In the case in which the semiconductor stack structure **10** has multiple first openings **16** and multiple second openings **17**, the contact area between the first electrode **31** and the first n-type semiconductor layer **11***c* at each of the first openings **16** is preferably larger than the contact area between the second electrode **32** and the second n-type semiconductor layer **12***c* at each of the second openings **17** in a

plan view.
[0061] In the light emitting element **100**, the areas of the first openings **16** in a plan view are the same. In the light emitting element **100**, the areas of the second openings **17** in a plan view are the same. The areas of the first openings **16** in a plan view may be different from one another. The areas of the second openings **17** in a plan view may be different from one another.
[0062] FIG. **3** is a cross-sectional view showing the current path in a first emission state of a light emitting element according to the first embodiment.
[0063] FIG. **4** is a cross-sectional view showing the current path in a second emission state of the light emitting element according to the first embodiment.
[0064] FIG. **5** is a cross-sectional view showing the current path in a third emission state of the light emitting element according to the first embodiment.
[0065] As shown in FIG. **3** to FIG. **5**, the light emitting element **100** is allowed to emit light in three emission states, i.e., the first emission state, the second emission state, and the third emission state.
[0066] The first emission state is a state in which both the first active layer **11***b* and the second active layer **12***b* are allowed to emit light. FIG. **3** shows the current path in the first emission state. As shown in FIG. **3**, in the first emission state, the first pad electrode **21** functions as the positive electrode and the lower electrode **23** functions as the negative electrode. An electric current flows from the first pad electrode **21** through the connection electrode **35**, the interlayer electrode **25**, the first p-type semiconductor layer **11***a*, the first active layer **11***b*, the first n-type semiconductor layer **11***c*, the intermediate layer **13**, the second p-type semiconductor later **12***a*, the second active layer **12***b*, the second n-type semiconductor layer **12***c*, and the second electrode **32** to the lower electrode **23**. This allows both the first active layer **11***b* and the second active layer **12***b* to emit light.
[0067] The second emission state is a state in which the first active layer **11***b* is allowed to emit light, but the second active layer **12***b* is not. FIG. **4** shows the current path in the second emission state. As shown in FIG. **4**, in the second emission state, the first pad electrode **21** functions as the positive electrode and the second pad electrode **22** functions as the negative electrode. An electric current flows from the first pad electrode **21** through the connection electrode **35**, the interlayer electrode **25**, the first p-type semiconductor layer **11***a*, the first active layer **11***b*, the first n-type semiconductor layer **11***c*, the first electrode **31**, and the connection electrode **35** to the second pad electrode **22**. This allows the first active layer **11***b* to emit light without allowing the second active layer **12***b* to emit light.
[0068] The third emission state is a state in which the second active layer **12***b* is allowed to emit light, but the first active layer **11***b* is not. FIG. **5** shows the current path in the third emission state. As shown in FIG. **5**, in the third emission state, the second pad electrode **22** functions as the positive electrode and the lower electrode **23** functions as the negative electrode. An electric current flows from the second pad electrode **22** through the connection electrode **35**, the first electrode **31**, the first n-type semiconductor layer **11***c*, the intermediate layer **13**, the second p-type semiconductor later **12***a*, the second active layer **12***b*, the second n-type semiconductor layer **12***c*, and the second electrode **32** to the lower electrode **23**. This allows the second active layer **12***b* to emit light without allowing the first active layer **11***b* to emit light.
[0069] In the case of providing negative electrodes that individually correspond to the first active layer **11***b* and the second active layer **12***b*, a large distance between the negative electrodes can cause the current density of the first active layer **11***b* to differ from that of the second active layer **12***b* to readily allow for light distribution nonuniformity. In the light emitting element **100**, the second openings **17** are arranged to overlap the first openings **16** in a plan view where the first electrode **31** is located in the first openings **16** and the second electrode **32** is located in the first openings **16** and the second openings **17**. This can reduce the distance between the first electrode **31**, which functions as the negative electrode when the first active layer **11***b* is allowed to emit light, and the second electrode **32**, which functions as the negative electrode when the second active later **12***b* is allowed to emit light, thereby reducing the current density variation between the

first active layer **11***b* and the second active layer **12***b*. This thus can reduce the light distribution variation between the first active layer **11***b* and the second active layer **12***b*.

[0070] In the case of allowing the second active layer **12***b* to emit light as in the third emission state, the intermediate layer **13** present in the current path tends to increase the forward voltage VF. In the case of a light emitting element **100** having a semiconductor stack structure **10** formed by successively depositing, on a growth substrate, a second n-type semiconductor layer **12***c*, a second active layer **12***b*, a second p-type semiconductor layer **12***a*, a first n-type semiconductor layer **11***c*, a first active layer **11***b*, and a first p-type semiconductor layer **11***a*, the crystallinity of the first active layer **11***b* tends to be inferior to that of the second active layer **12***b*. For this reason, when the first active layer **11***b* is allowed to emit light as in the second emission state, the forward voltage VF readily becomes higher than that in the case in which the second active layer **12***b* is allowed to emit light as in the third emission state. Accordingly, in the light emitting element **100**, setting the contact area between the first electrode **31** and the first n-type semiconductor layer **11***c* in a first opening **16** larger than the contact area between the second electrode **32** and the second n-type semiconductor layer **12***c* in a second opening **17** can reduce the forward voltage VF in the first electrode **31** thereby improving the light intensity in the second and third emission states.

[0071] In the light emitting element **100**, moreover, each second connection part **32***a* is surrounded by a first connection part **31***a*. This can bring the in-plane light distribution property of the first active layer **11***b* close to that of the second active layer **12***b* as compared to the case of locating the first openings **16** and the second openings **17** so as not to overlap and providing the first connection parts **31***a* and the second connection parts **32***a*, respectively.

[0072] In the light emitting element **100**, furthermore, the light emission of first active layer **11***b* and the light emission of the second active layer **12***b* can be controlled individually. Accordingly, in the case in which the peak wavelength of the light from the first active layer **11***b* differs from the peak wavelength of the light from the second active layer **12***b*, the peak wavelength of the light from the light emitting element can be switched by switching the active layer that is allowed to emit light. For example, in the case in which the peak wavelength of the light from the first active layer **11***b* is the same as the peak wavelength of the light from the second active layer **12***b*, the active layer **11***b* and the second active layer **12***b* can be used alternately by switching the active layer that is allowed to emit light. This can reduce the impact of the heat on the first active layer **11***b* and the second active layer **12***b* thereby reducing the shifting of the peak wavelength attributable to the heat generated by the semiconductor stack structure **10** as compared to the case in which a single active layer is allowed to emit light. For example, in the case in which the first active layer **11***b* and the second active layer **12***b* emit ultraviolet light, the first active layer **11***b* and the second active layer **12***b* are easily affected by the heat. Thus, switching between the active layers for light emission can be particularly effective in reducing the impact of the heat on the active layers.

Variation

[0073] FIG. **6** is a plan view showing a light emitting element according to a variation of the first embodiment.

[0074] FIG. **7** is a cross-sectional view showing the light emitting element according to the variation of the first embodiment.

[0075] FIG. **7** is a cross section taken along line VII-VII in FIG. **6**.

[0076] As shown in FIG. **6** and FIG. **7**, in a light emitting element **100**A according to the variation of the first embodiment, the shapes of the first openings **16** and the second openings **17** and the positions of the first connection parts **31***a* and the second connection parts **32***a* differ from those of the light emitting element **100**. The light emitting element **100**A is the same as the light emitting element **100** otherwise.

[0077] In the light emitting element **100**A, the plan view shape of each second opening **17** is a semicircle. In the light emitting element **100**A, moreover, the second connection parts **32***a* are not surrounded by the first connection parts **31***a*.

[0078] In the light emitting element **100**A, similar to the light emitting element **100**, the second opening **17** is positioned to overlap the first opening **16**, the first electrode **31** is located in the first opening **16**, and the second electrode **32** is located in the first opening **16** and the second opening **17** to thereby reduce the light distribution variation between the first active layer **11***b* and the second active layer **12***b*.

Second Embodiment

[0079] FIG. **8** is a plan view showing a light emitting element according to a second embodiment.

[0080] FIG. **9** is a cross-sectional view showing the light emitting element according to the second embodiment.

[0081] FIG. **9** is a cross section taken along line IX-IX in FIG. **8**.

[0082] As shown in FIG. **8** and FIG. **9**, a light emitting element **200** according to the second embodiment primarily differs from the light emitting element **100** such that the semiconductor stack structure **10** further includes third openings **18** and a third electrode **33**. The light emitting element **200** essentially has the same structure as the light emitting element **100** described above otherwise.

[0083] As shown in FIG. **8** and FIG. **9**, the semiconductor stack structure **10** has a first region **10***x* and a second region **10***y* in a plan view. The second region **10***y* does not overlap the first region **10***x* in the plan view. A first opening **16** and a second opening **17** are located in the first region **10***x*. A third opening **18** is located in the second region **10***y*. The third opening **18** does not overlap the first opening **16** and the second opening **17** in the plan view.

[0084] A third opening **18** is provided continuously in the first p-type semiconductor layer **11***a* and the first active layer **11***b*. The third opening **18** is a hole that passes through the first p-type semiconductor layer **11***a* and the first active layer **11***b* in the Z direction. In the light emitting element **200**, the third opening **18** is provided continuously in the first p-type semiconductor layer **11***a*, the first active layer **11***b*, and the first n-type semiconductor layer **11***c*. In the light emitting element **200**, the third opening **18** is a hole that passes through the first p-type semiconductor layer **11***a* and the first active layer **11***b* before reaching the first n-type semiconductor layer **11***c* in the Z direction. The third opening **18** does not pass through the first n-type semiconductor layer **11***c*. In the light emitting element **200**, the semiconductor stack structure **10** has multiple third openings **18**. In the light emitting element **200**, the semiconductor stack structure **10** has at least one third opening **18**.

[0085] In the light emitting element **200**, the plan view shape of a third opening **18** is a perfect circle. The plan view shape of a third opening **18** may be a semicircle, ellipse, polygon, or the like.

[0086] The third electrode **33** electrically connects the second pad electrode **22** and the first n-type semiconductor layer **11***c*. The third electrode **33** is located in a third opening **18**. The third electrode **33** is connected to the first electrode **31**. The third electrode **33** is essentially the same as the first electrode **31** except for the location which is in the third opening **18**. In other words, the third electrode **33** is an electrode that corresponds to the first electrode **31** in a second region **10***y*. In the light emitting element **200**, the third electrode **33** and the first n-type semiconductor layer **11***c* are electrically connected by being in contact with one another at a third connection part **33***a*. The material for the third electrode **33** may be the same as that for the first electrode **31**.

[0087] As shown in FIG. **8**, in the plan view, the semiconductor stack structure **10** has one central region **10***a*, two intermediate regions **10***b*, and two outer regions **10***c*. The central region **10***a* in the plan view is located halfway between the first pad electrode **21** and the second pad electrode **22**. The central region **10***a* includes the center of the semiconductor stack structure **10** in the plan view, for example. One of the intermediate regions **10***b* is located between the central region **10***a* and the first pad electrode **21**, and the other between the central region **10***a* and the second pad electrode **22**. One of the outer regions **10***c* is located between one of the intermediate regions **10***b* and the first pad electrode **21**, and the other between the other intermediate region **10***b* and the second pad electrode **22**.

[0088] In the plan view, the portion of the diagonal line DG of the light emitting element **200** passing through the first pad electrode **21** and the second pad electrode **22** that overlaps the semiconductor stack structure **10** is designated as the first line section LS. The four straight lines orthogonal to the first line section LS are designated as the first line IL**1**, second line IL**2**, third line IL**3**, and fourth line IL**4** from the first pad electrode **21** side. The first line section LS is equally divided into five sections by the first line IL**1**, the second line IL**2**, the third line IL**3**, and the fourth line IL**4**. The central region **10***a* is the region of the semiconductor stack structure **10** that is located between the second line IL**2** and the third line IL**3** in the plan view. The intermediate regions **10***b* are the region of the semiconductor stack structure **10** located between the first line IL**1** and the second line IL**2** and the region of the semiconductor stack structure **10** located between the third line IL**3** and the fourth line IL**4** in the plan view. One of the outer regions **10***c* is the region of the semiconductor stack structure **10** located between the first line ILI and the outer edge **10***e* of the semiconductor stack structure **10**, and the other one of the outer regions **10***c* is the region of the semiconductor stack structure **10** located between the fourth line IL**4** and the outer edge **10***e* of the semiconductor stack structure **10** in the plan view. The outer regions **10***c* are positioned outward from the intermediate regions **10***b* using the center of the semiconductor stack structure **10** as a reference.

[0089] A second region **10***y* is located in the central region **10***a*, for example. In the light emitting element **200**, the second regions **10***y* are located in the central region **10***a* and not located in the intermediate regions **10***b* and the outer regions **10***c*. In other words, in the light emitting element **200**, the third electrode **33** is in contact with the first n-type semiconductor layer **11***c* in the central region **10***a*, and not in contact with the first n-type semiconductor layer **11***c* in the intermediate regions **10***b* and the outer regions **10***c*. In the light emitting element **200**, moreover, the first regions **10***x* are located in the central region **10***a*, the intermediate regions **10***b*, and the outer regions **10***c*. In other words, in the light emitting element **200**, the first electrode **31** is in contact with the first n-type semiconductor layer **11***c* in the central region **10***a*, the intermediate regions **10***b*, and the outer regions **10***c*. In the light emitting element **200**, the second electrode **32** is in contact with the second n-type semiconductor layer **12***c* in the central region **10***a*, the intermediate regions **10***b*, and the outer regions **10***c*.

[0090] In the light emitting element **200**, the first openings **16** and the third openings **18** are alternately arranged in the central region **10***a*. The layout of the first openings **16** and the third openings **18** is not limited to this. For example, the layout may be such that two or more third openings **18** may be provided between two first openings **16**, or one third opening **18** between two first openings **16** in part while two or more third openings **18** between two first openings **16** in part.

[0091] In the light emitting element **200**, the second openings **17** and the third openings **18** are alternately arranged in the central region **10***a*. The layout of the second openings **17** and the third openings **18** is not limited to this. For example, the layout may be such that two or more third openings **18** may be provided between two second openings **17**, or one third opening **18** between two second openings **17** in part while two or more third openings **18** between two second openings **17** in part.

[0092] As described above, in the light emitting element **200**, the first openings **16** and the second openings **17** are located in the first regions **10***x*, the third openings **18** are located in the second regions **10***y* which do not overlap the first regions **10***x* in a plan view, and the third electrode **33** is located in the third openings **18**. In the case of a light emitting element **200** having a semiconductor stack structure formed by successively depositing, on a growth substrate, a second n-type semiconductor layer **12***c*, a second active layer **12***b*, a second p-type semiconductor layer **12***a*, a first n-type semiconductor layer **11***c*, a first active layer **11***b*, and a first p-type semiconductor layer **11***a*, the first p-type semiconductor layer **11***a*, the first active layer **11***b*, and the first n-type semiconductor layer **11***c* tend to have inferior crystallinity to the second p-type semiconductor layer

12*a*, the second active layer **12***b*, and the second n-type semiconductor layer **12***c*, and are less bright. Accordingly, placing the third electrode **33** that is in contact with the first n-type semiconductor layer **11***c* separately from the first electrode **31** can increase the contact areas between the first n-type semiconductor layer **11***c* and the electrodes, thereby improving the light intensity when allowing the first active layer **11***b* to emit light.

[0093] The central region **10***a* that is located farther away from the first pad electrode **21** and the second pad electrode **22** than the intermediate regions **10***b* and the outer regions **10***c* are tends to have a lower current density than the intermediate regions **10***b* and the outer regions **10***c*. Accordingly, locating the third openings **18** (the third electrode **33**) in the second regions **10***y* located in the central region **10***a* can improve the light intensity in the central region **10***a* which readily has a lower current density than the intermediate regions **10***b* and the outer regions **10***c*. This can reduce the in-plane light distribution nonuniformity of the semiconductor stack structure **10**.

Variation

[0094] FIG. **10** is a plan view of a light emitting element according to a variation of the second embodiment.

[0095] As shown in FIG. **10**, a light emitting element **200**A according to the variation of the second embodiment primarily differs from the light emitting element **200** in terms of the layout of the first openings **16**, the second openings **17**, and the third openings **18** in the plan view. The light emitting element **200**A has practically the same structure as that of the light emitting element **200** described above otherwise.

[0096] In the light emitting element **200**A, the second regions **10***y* are located in the central region **10***a*, and not in the intermediate regions **10***b* and the outer regions **10***c*. In other words, in the light emitting element **200**A, the third electrode **33** is in contact with the first n-type semiconductor layer **11***c* in the central region **10***a*, and not in contact with the first n-type semiconductor layer **11***c* in the intermediate regions **10***b* and the outer regions **10***c*. In the light emitting element **200**A, the first regions **10***x* are located in the intermediate regions **10***b* and the outer regions **10***c*, and not located in the central region **10***a*. In other words, in the light emitting element **200**A, the first electrode **31** is in contact with the first n-type semiconductor layer **11***c* in the intermediate regions **10***b* and the outer regions **10***c*, and not in contact with the first n-type semiconductor layer **11***c* in the central region **10***a*. In the light emitting element **200**A, the second electrode **32** is in contact with the second n-type semiconductor layer **12***c* in the intermediate regions **10***b* and the outer regions **10***c*, and not in contact with the second n-type semiconductor layer **12***c* in the central region **10***a*.

[0097] In the case of a light emitting element **200**A having a semiconductor stack structure formed by successively depositing, on a growth substrate, a second n-type semiconductor layer **12***c*, a second active layer **12***b*, a second p-type semiconductor layer **12***a*, a first n-type semiconductor layer **11***c*, a first active layer **11***b*, and a first p-type semiconductor layer **11***a*, the light intensity when allowing the first active layer **11***b* to emit light can be similarly improved by locating the third electrode **33** in the third openings **18**.

Third Embodiment

[0098] FIG. **11** is a plan view showing a light emitting element according to a third embodiment.

[0099] As shown in FIG. **11**, in the light emitting element **300** according to the third embodiment, the plan view areas of the first openings **16** and the second openings **17** are different from those in the light emitting element **100**. The light emitting element **300** has essentially the same structure as the light emitting element **100** described above otherwise.

[0100] As shown in FIG. **11**, in the plan view, the semiconductor stack structure **10** has a central region **10***a*, intermediate regions **10***b*, and outer regions **10***c*. The positional relationship of the central region **10***a*, the intermediate regions **10***b*, and the outer regions **10***c* is the same as that of the central region **10***a*, the intermediate regions **10***b*, and the outer regions **10***c* in the light emitting element **200** described above.

[0101] The first openings **16** include multiple first central openings **16***a*, multiple first intermediate openings **16***b*, and multiple first outer openings **16***c*. The first central openings **16***a* are located in the central region **10***a*. The first intermediate openings **16***b* are located in the intermediate regions **10***b*. The first outer openings **16***c* are located in the outer regions **10***c*.
[0102] The second openings **17** include multiple second central openings **17***a*, multiple second intermediate openings **17***b*, and multiple second outer openings **17***c*. The second central openings **17***a* are located in the central region **10***a*. The second intermediate openings **17***b* are located in the intermediate regions **10***b*, and the second outer openings **17***c* are located in the outer regions **10***c*.
[0103] In the light emitting element **300**, each of the first central openings **16***a* is larger in area than each of the first intermediate openings **16***b* in a plan view.
[0104] Moreover, each of the first intermediate openings **16***b* is larger in area than each of the first outer openings **16***c* in a plan view.
[0105] In the light emitting element **300**, the plan view areas of the first central openings **16***a* are the same. In the light emitting element **300**, the plan view areas of the first intermediate openings **16***b* are the same. In the light emitting element **300**, the plan view areas of the first outer openings **16***c* are the same. The plan view areas of the first central openings **16***a* may be different from one another. The plan view areas of the first intermediate openings **16***b* may be different from one another. The plan view areas of the first outer openings **16***c* may be different from one another.
[0106] Here, the sum of the plan view areas where the first electrode **31** is in contact with the first n-type semiconductor layer **11***c* at the first central openings **16***a* is designated as the total area S.sub.SUMa. The sum of the plan view areas where the first electrode **31** is in contact with the first n-type semiconductor layer **11***c* at the first intermediate openings **16***b* is designated as the total area S.sub.SUMb. The sum of the plan view areas where the first electrode **31** is in contact with the first n-type semiconductor layer **11***c* at the first outer openings **16***c* is designated as the total area S.sub.SUMc. In the light emitting element **300**, the total area S.sub.SUMa is larger than the total area S.sub.SUMb (S.sub.SUMa>S.sub.SUMb). The total area S.sub.SUMb is larger than the total area S.sub.SUMc (S.sub.SUMb>S.sub.SUMc).
[0107] In the light emitting element **300**, the plan view area of each of the second central openings **17***a* is larger than the plan view area of each of the second intermediate openings **17***b*. The plan view area of each of the second intermediate openings **17***b* is larger than the plan view area of each of the second outer openings **17***c*.
[0108] In the light emitting element **300**, the plan view areas of the second central openings **17***a* are the same. In the light emitting element **300**, the plan view areas of the second intermediate openings **17***b* are the same. In the light emitting element **300**, the plan view areas of the second outer openings **17***c* are the same. The plan view areas of the second central openings **17***a* may be different from one another. The plan view areas of the second intermediate openings **17***b* may be different from one another. The plan view areas of the second outer openings **17***c* may be different from one another.
[0109] Here, the area of contact between the second electrode **32** and the second n-type semiconductor layer **12***c* at one of the second central openings **17***a* in a plan view is designated as an area S**2***a*. The area of contact between the first electrode **31** and the first n-type semiconductor layer **11***c* at one of the first central openings **16***a* in a plan view is designated as an area S**1***a*. The area of contact between the second electrode **32** and the second n-type semiconductor layer **12***c* at one of the second intermediate openings **17***b* in a plan view is designated as an area S**2***b*. The area of contact between the first electrode **31** and the first n-type semiconductor layer **11***c* at one of the first intermediate openings **16***b* in a plan view is designated as an area S**1***b*. The area of contact between the second electrode **32** and the second n-type semiconductor layer **12***c* at one of the second outer openings **17***c* in a plan view is designated as an area S**2***c*. The area of contact between the first electrode **31** and the first n-type semiconductor layer **11***c* at one of the first outer openings **16***c* in a plan view is designated as an area S**1***c*. In the light emitting element **300**, the ratio of the

area S**1***a* to the area S**2***a* (S**1***a*/S**2***a*) is the same as the ratio of the area S**1***b* to the area S**2***b* (S**1***b*/S**2***b*) and the ratio of the area S**1***c* to the area S**2***c* (S**1***c*/S**2***c*) (S**1***a*/S**2***a*=S**1***b*/S**2***b*=S**1***c*/S**2***c*).
[0110] The central region **10***a* tends to have a lower current density than the intermediate regions **10***b*. The intermediate regions **10***b* tend to have a lower current tensity than the outer regions **10***c*. Accordingly, in the light emitting element **300**, the sum of the areas of contact between the first electrode **31** and the first n-type semiconductor layer **11***c* at the first openings **16** located in the central region **10***a* (first central openings **16***a*) (total area S.sub.SUMa) is set to be larger than the sum of the areas of contact between the first electrode **31** and the first n-type semiconductor layer **11***c* at the first openings **16** located in the intermediate region **10***b* (first intermediate openings **16***b*) (total area S.sub.SUMb), and the total area S.sub.SUMb is set to be larger than the sum of the areas of contact between the first electrode **31** and the first n-type semiconductor layer **11***c* at the first openings **16** located in the outer regions **10***c* (first outer openings **16***c*) (total area S.sub.SUMc). This can achieve a high current density in the intermediate regions **10***b* where the current density tends to be lower than the outer regions **10***c*, while achieving an even higher current density in the central region **10***a* which tends to have a lower current density than the intermediate regions **10***b*. This can reduce the in-plane light distribution nonuniformity of the semiconductor stack structure **10**.
[0111] In the light emitting element **300**, moreover, the electrode and semiconductor layer contact area ratio (S**1***a*/S**2***a*) at each opening in the central region **10***a*, the electrode and semiconductor layer contact area ratio (S**1***b*/S**2***b*) at each opening in the intermediate regions **10***b*, and the electrode and semiconductor layer contact area ratio (S**1***c*/S**2***c*) at each opening in the outer regions **10***c* are set to be the same. This can reduce the current density nonuniformity in the openings, thereby reducing the degradation of semiconductor layers.
Variation
[0112] FIG. **12** is a plan view of a light emitting element according to a variation of the third embodiment.
[0113] As shown in FIG. **12**, a light emitting element **300**A according to the variation of the third embodiment primarily differs from the light emitting element **300** in terms of the sizes of the areas of the first openings **16** and the second openings **17** in a plan view. The light emitting element **300**A has essentially the same structure as the light emitting element **300** described above otherwise.
[0114] In the light emitting element **300**A, similar to the light emitting element **300** described above, the first central openings **16***a* are larger in area than the first intermediate openings **16***b* in a plan view. Moreover, the first intermediate openings **16***b* are larger in area than the first outer openings **16***c* in a plan view.
[0115] In the light emitting element **300**A, the plan view area of each of the second central openings **17***a* is the same as the plan view area of each of the second intermediate openings **17***b* and the second outer openings **17***c*.
[0116] In the light emitting element **300**A, the ratio of the area S**1***a* to the area S**2***a* (S**1***a*/S**2***a*) is higher than the ratio of the area S**1***b* to the area S**2***b* (S**1***b*/S**2***b*) (S**1***a*/S**2***a*>S**1***b*/S**2***b*). Furthermore, the ratio of the area S**1***b* to the area S**2***b* (S**1***b*/S**2***b*) is higher than the ratio of the area S**1***c* to S**2***c* (S**1***c*/S**2***c*) (S**1***b*/S**2***b*>S**1***c*/S**2***c*).
[0117] In the light emitting element **300**A, similar to the light emitting element **300**, the total area S.sub.SUMa is larger than the total area S.sub.SUMb (S.sub.SUMa>S.sub.SUMb). Also, the total area S.sub.SUMb is larger than the total are S.sub.SUMc (S.sub.SUMb>S.sub.SUMc).
[0118] In the light emitting element **300**A, the plan view areas of the first central openings **16***a* are the same. In the light emitting element **300**A, the plan view areas of the first intermediate openings **16***b* are the same. In the light emitting element **300**A, the plan view areas of the first outer openings **16***c* are the same. The plan view areas of the first central openings **16***a* may be different from one another. The plan view areas of the first intermediate openings **16***b* may be different from one

another. The plan view areas of the first outer openings **16**_c_ may be different from one another.

[0119] In the light emitting element **300**A, the plan view areas of the second central openings **17**_a_ are the same. In the light emitting element **300**A, the plan view areas of the second intermediate openings **17**_b_ are the same. In the light emitting element **300**A, the plan view areas of the second outer openings **17**_c_ are the same. The plan view areas of the second central openings **17**_a_ may be different from one another. The plan view areas of the second intermediate openings **17**_b_ may be different from one another. The plan view areas of the second outer openings **17**_c_ may be different from one another.

[0120] In the light emitting element **300**A, the sum of the areas of contact between the first electrode **31** and the first n-type semiconductor layer **11**_c_ at the first openings **16** located in the central region **10**_a_ (first central openings **16**_a_) (total area S.sub.SUMa) is set to be larger than the sum of the areas of contact between the first electrode **31** and the first n-type semiconductor layer **11**_c_ at the first openings **16** located in the intermediate region **10**_b_ (first intermediate openings **16**_b_) (total area S.sub.SUMb). The total area S.sub.SUMb is set to be larger than the sum of the areas of contact between the first electrode **31** and the first n-type semiconductor layer **11**_c_ at the first openings **16** located in the outer regions **10**_c_ (first outer openings **16**_c_) (total area S.sub.SUMc). This can reduce the in-plane light distribution nonuniformity of the semiconductor stack structure **10**.

[0121] In the light emitting element **300**A, moreover, the electrode and semiconductor layer contact area ratio (S**1**_a_/S**2**_a_) at each opening in the central region **10**_a_ is set to be higher than the electrode and semiconductor layer contact area ratio (S**1**_b_/S**2**_b_) at each opening in the intermediate regions **10**_b_. The electrode and semiconductor layer contact area ratio (S**1**_b_/S**2**_b_) at each opening in the intermediate regions **10**_b_ is set to be higher than the electrode and semiconductor layer contact area ratio (S**1**_c_/S**2**_c_) at each opening in the outer regions **10**_c_. This can reduce the in-plane current density nonuniformity of the semiconductor stack structure **10**.

Manufacturing Method

[0122] FIG. **13** to FIG. **17** are cross-sectional views each showing a part of a method of manufacturing a light emitting element according to the first embodiment.

[0123] In the method of manufacturing a light emitting element **100**, a semiconductor stack structure **10** having multiple first openings **16** is formed on a growth substrate **80** first. Then on the semiconductor stack structure **10**, an interlayer electrode **25**, a connection electrode **35**, and an interlayer insulation film **50** are formed. Then a first electrode **31** is formed on the interlayer insulation film **50** and in the first openings **16**. By following these steps, a structure having a first electrode **31** located in the first openings **16** is provided as shown in FIG. **13**.

[0124] Then as shown in FIG. **14**, second openings **17** are formed by partially removing the first electrode **31** and the semiconductor stack structure **10** located in the first openings **16**.

[0125] Then as shown in FIG. **15**, an insulation layer **45** is formed on the first electrode **31** and in the second openings **17**.

[0126] Then as shown in FIG. **16**, a portion of the semiconductor stack structure **10** is exposed by partially removing the insulation layer **45**.

[0127] Then as shown in FIG. **17**, a second electrode **32** is formed on the exposed semiconductor stack structure **10** and the insulation layer **45**. Then on the second electrode **32**, a lower electrode **23** is formed.

[0128] Embodiments of the present disclosure may include those described below.

[0129] As described above, according to an embodiment of the present disclosure, a light emitting element capable of reducing light distribution variation between the first active layer and the second active layer can be provided.

[0130] The embodiments described above are examples that give shape to the present invention, but the present invention is not limited to these embodiments. For example, embodiments resulting from adding to, removing from, or modifying certain constituent elements or processes in the embodiments described above are also encompassed by the present invention. The embodiments described above, moreover, can be implemented in combination with one another.

Claims

1. A light emitting element comprising: a semiconductor stack structure comprising: a first p-type semiconductor layer, a first active layer disposed on the first p-type semiconductor layer, a first n-type semiconductor layer disposed on the first active layer, an intermediate layer disposed on the first n-type semiconductor layer, a second p-type semiconductor layer disposed on the intermediate layer, a second active layer disposed on the second p-type semiconductor layer, and a second n-type semiconductor layer disposed on the second active layer, wherein: the semiconductor stack structure includes a first opening provided continuously in the first p-type semiconductor layer and the first active layer, and a second opening located to overlap the first opening in a plan view and provided continuously in the first n-type semiconductor layer, the intermediate layer, the second p-type semiconductor layer, and the second active layer; a first electrode located in the first opening and electrically connected to the first n-type semiconductor layer; a second electrode located in the first opening and the second opening and electrically connected to the second n-type semiconductor layer; and an insulation layer disposed between and in contact with the first electrode and the second electrode.
2. The light emitting element according to claim 1, wherein a plan view area of contact between the first electrode and the first n-type semiconductor layer at the first opening is larger than a plan view area of contact between the second electrode and the second n-type semiconductor layer at the second opening.
3. The light emitting element according to claim 1, wherein the second connection part where the second electrode is in contact with the second n-type semiconductor layer at the second opening is surrounded by the first connection part where the first electrode is in contact with the first n-type semiconductor layer at the first opening.
4. The light emitting element according to claim 1, wherein: the semiconductor stack structure comprises a first region and a second region not overlapping the first region in the plan view; the first opening and the second opening are positioned in the first region; the semiconductor stack structure further includes a third opening positioned in the second region and provided continuously in the first p-type semiconductor layer and the first active layer; and the light emitting element further comprises a third electrode located in the third opening and electrically connected to the first n-type semiconductor layer.
5. The light emitting element according to claim 4, further comprising a first pad electrode electrically connected to the first p-type semiconductor layer; and a second pad electrode electrically connected to the second n-type semiconductor layer; wherein: the semiconductor stack structure comprises: a central region positioned halfway between the first pad electrode and the second pad electrode in the plan view, a first intermediate region positioned between the central region and the first pad electrode, and a second intermediate region positioned between the central region and the second pad electrode, in the plan view, and a first outer region positioned between one of the intermediate regions and the first pad electrode, and a second outer region positioned between the other of the intermediate region and the second pad electrode, in the plan view; and the second region is positioned in the central region.
6. The light emitting element according to claim 1, further comprising: a first pad electrode electrically connected to the first p-type semiconductor layer; and a second pad electrode electrically connected to the second n-type semiconductor layer; wherein: the semiconductor stack structure comprises: a central region positioned halfway between the first pad electrode and the second pad electrode in the plan view, a first intermediate region positioned between the central region and the first pad electrode, and a second intermediate region positioned between the central

region and the second pad electrode, in the plan view, a first outer region positioned between one of the intermediate regions and the first pad electrode, and a second outer region positioned between the other of the intermediate regions and the second pad electrode, in the plan view; the semiconductor stack structure includes a plurality of the first openings, which include: a plurality of first central openings located in the central region, a plurality of first intermediate openings located in the intermediate regions, and a plurality of first outer openings located in the outer regions; a sum of plan view areas where the first electrode is in contact with the first n-type semiconductor layer at the first central openings is larger than a sum of plan view areas where the first electrode is in contact with the first n-type semiconductor layer at the first intermediate openings; and a sum of the plan view areas where the first electrode is in contact with the first n-type semiconductor layer at the first intermediate openings is larger than a sum of plan view areas where the first electrode is in contact with the first n-type semiconductor layer at the first outer openings.

7. The light emitting element according to claim 6, wherein: the semiconductor stack structure includes a plurality of the second openings, which include: a plurality of second central openings located in the central region, a plurality of second intermediate openings located in the intermediate regions, and a plurality of second outer openings located in the outer regions; and a ratio of a plan view area of contact between the first electrode and the first n-type semiconductor layer at one of the first central openings to a plan view area of contact between the second electrode and the second n-type semiconductor layer at one of the second central openings is the same as a ratio of a plan view area of contact between the first electrode and the first n-type semiconductor layer at one of the first intermediate openings to a plan view area of contact between the second electrode and the second n-type semiconductor layer at one of the second intermediate openings and a ratio of a plan view area of contact between the first electrode and the first n-type semiconductor layer at one of the first outer openings to a plan view area of contact between the second electrode and the second n-type semiconductor layer at one of the second outer openings.

8. The light emitting element according to claim 6, wherein: the semiconductor stack structure includes a plurality of the second openings, which include: a plurality of second central openings located in the central region, a plurality of second intermediate openings located in the intermediate regions, and a plurality of second outer openings located in the outer regions; and a ratio of a plan view area of contact between the first electrode and the first n-type semiconductor layer at one of the first central openings to a plan view area of contact between the second electrode and the second n-type semiconductor layer at one of the second central openings is higher than a ratio of a plan view area of contact between the first electrode and the first n-type semiconductor layer at one of the first intermediate openings to a plan view area of contact between the second electrode and the second n-type semiconductor layer at one of the second intermediate openings; and a ratio of a plan view area of contact between the first electrode and the first n-type semiconductor layer at one of the first intermediate openings to a plan view area of contact between the second electrode and the second n-type semiconductor layer at one of the second intermediate openings is higher than a ratio of a plan view area of contact between the first electrode and the first n-type semiconductor layer at one of the first outer openings to a plan view area of contact between the second electrode and the second n-type semiconductor layer at one of the second outer openings.

9. The light emitting element according to claim 1, wherein an area of the second active layer is larger than an area of the first active layer in the plan view.